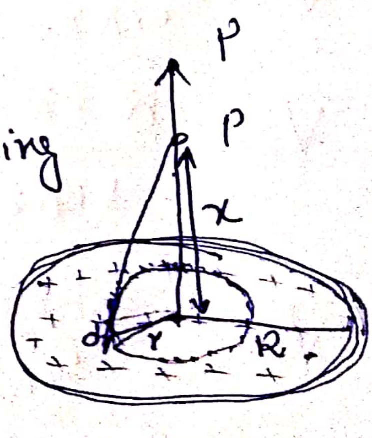
* ELECTRI POTENTIAL DUE TO UNIFORMLY CHARGED DISK :-

- Let us consider a Disk of Radius 'R' is uniformly distributed by a charge '+q' such that surface charge Density is 'σ'
- We want to calculate Electric potential at point 'P' which is at Distance 'x' from the center of Disk
- To calculate potential at poin 'P' first we consider a Ring of radius 'r' such that 'r<R' and calculate Potential at point 'P' due to ring Then we will Integrate it for the Potential for DISK.
- As we have already calculated E.P for Ring which is given as

$$V = \frac{1}{4\pi\varepsilon_0} \frac{q}{\sqrt{x^2+r^2}}$$

r = radius of Ring

R = " of Disk



Surface charge Density

$$\sigma = \frac{dq}{dA}$$

$$dq = \sigma \, dA$$

$$q = \sigma \int dA$$

$$q = \sigma \, 2\pi r . dr$$

$$\boxed{dV = \frac{1}{4\pi\varepsilon_0}\frac{q}{\sqrt{x^2+r^2}}}$$

$q = \sigma(2\pi r)dr$ ↳ width ↳ C.F

$$dV = \frac{1}{4\pi}\frac{\sigma 2\pi r\,dr}{\sqrt{x^2+r^2}}$$

This is potential at point 'P' due to ring

Now, for Total potential

$$\int dV = \int_0^R \frac{1}{4\pi\varepsilon_0}\frac{\sigma 2\pi r\,dr}{\sqrt{x^2+r^2}}$$

$$\int dV = \frac{1}{4\pi\varepsilon_0}(\sigma.2\pi)\int_0^R \frac{r\,dr}{\sqrt{x^2+r^2}} \quad \text{— (A)}$$

using values from Integration

Integration by substitution

let

$$\boxed{x^2+r^2=t^2}$$

Diff w.r.t

$$\frac{d}{dt}(x^2)+\frac{d}{dt}(r^2)=\frac{d}{dt}(t^2)$$

$$0+2r\frac{dr}{dt}=2t$$

$$r\,dr = \frac{\cancel{2}t\times dt}{\cancel{2}}$$

$$\boxed{r\,dr = t\,dt}$$

$$V = \frac{\sigma 2\pi}{4\pi\varepsilon_0}\int_0^R \frac{t\,dt}{\sqrt{t^2}}$$

$$V = \frac{\sigma 2\pi}{4\pi\varepsilon_0}\int_0^R dt$$

$$V = \frac{\sigma 2\pi}{4\pi\varepsilon_0}[t]_0^R$$

$$V = \frac{\sigma 2\pi}{4\pi\varepsilon_0}\left[\sqrt{x^2+r^2}\right]_0^R$$

$$V = \frac{\sigma \cancel{2}\pi}{{}_2\cancel{4}\pi\varepsilon_0}\left[\sqrt{x^2+R^2} - \sqrt{x^2+0}\right]$$

$$V = \frac{\sigma \pi}{2\pi\varepsilon_0}\left[\sqrt{x^2+R^2} - x\right]$$ final expression

* Maximum potential

as $R \to \infty$, $V \to \infty$

↳ Infinite Sheet ↳ Infinite potential

* For $x = 0$

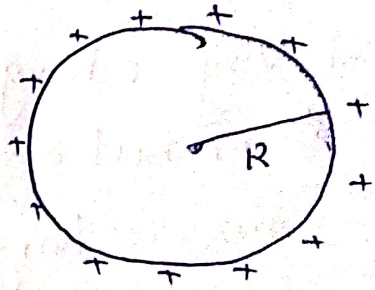
$$\boxed{V = \frac{\sigma}{2\varepsilon_0} R}$$

ii) ELECTRIC POTENTIAL FOR A CHARGED CONDUCTING SPHERE

Consider a charge is uniformly Distributed over a conducting sphere of Radius 'R'

* Electric field Inside sphere

$E_{inside} = 0$ bcz $q = 0$



* Electric Field outside sphere

$E_{out} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2}$ when $r > R$

* Potential outside sphere

$V_{out} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r}$ when $r > R$

* Potential at surface

$V_{surface} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{R}$ $(r = R)$

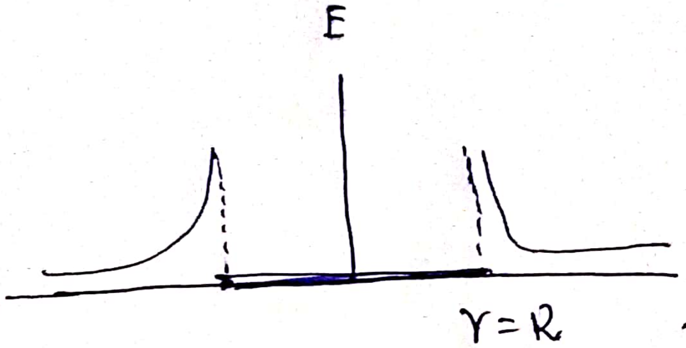
* Potentail Inside sphere

$V_{inside} = \frac{1}{4\pi\varepsilon_0} \frac{Q}{R}$ ∵ No any work is Required to move a test Charge,, ④

* Graph

Electric Field V/s Distance

$E = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r^2}$

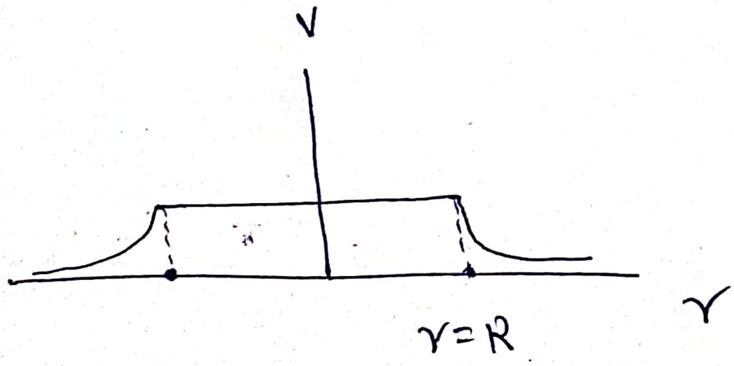


→ E.F is zero Inside the sphere (r=R)

→ E.F Decreases as move away from surface of sphere r>R

* ## Electric Potential V/s Distance

$V = \frac{1}{4\pi\varepsilon_0} \frac{Q}{r}$



→ potential is constat Inside the sphere

→ potential will Decrease as we move away from Surface i.e r>R